

ARE, Sweden

WMO: 022170

Lat: 63.3738N

Lon: 13.1607E

Elev: 1322

StdP: 14.01

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-6.0	-16.4	2.4	-12.4	-9.8	3.4	-6.1	22.4	33.7	20.2	32.2	0.5	320	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	77.9	62.9	74.1	61.7	70.1	59.3	64.5	74.9	62.6	71.3	60.6	68.4	5.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	82.9	68.0	58.8	77.7	66.1	57.1	72.9	64.3	30.2	75.0	28.8	71.2	27.4	68.6	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	15.8	13.5	-17.8	83.0	6.1	2.9	-22.2	85.1	-25.8	86.8	-29.2	88.5	-33.6	90.6		
(5)			-17.7	66.8	5.4	2.0	-21.6	68.2	-24.8	69.4	-27.8	70.5	-31.7	72.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	19.4	21.7	27.7	36.6	44.6	53.0	58.3	55.9	47.3	37.6	28.5	21.9		
	DBStd	15.77	12.25	11.09	8.59	6.10	6.38	6.71	5.88	5.50	5.71	6.92	8.37	11.91		
	HDD50	5104	948	793	692	402	194	42	4	11	116	385	645	872		
	HDD65	9948	1413	1213	1157	852	632	362	223	286	530	848	1095	1337		
	CDD50	651	0	0	0	0	27	133	260	193	36	2	0	0		
	CDD65	20	0	0	0	0	0	3	14	3	0	0	0	0		
	CDH74	314	0	0	0	0	6	88	165	55	0	0	0	0		
	CDH80	32	0	0	0	0	0	3	27	2	0	0	0	0		
	WSAvg	5.3	5.1	5.6	6.1	5.6	6.1	5.9	4.9	4.2	4.8	5.4	5.0	4.9		
	PrecAvg	25.0	1.9	1.5	1.5	1.1	1.6	2.8	3.6	3.3	2.1	2.0	1.8	1.6		
(15) Precipitation	PrecMax	33.9	3.8	4.1	2.9	2.5	3.4	4.9	7.0	5.6	4.3	5.3	3.7	3.4		
	PrecMin	18.7	0.1	0.2	0.6	0.2	0.7	1.1	0.7	1.5	0.5	0.5	0.3	0.7		
	PrecStd	4.0	0.9	0.8	0.6	0.6	0.7	1.0	1.8	1.4	1.1	1.1	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	42.0	48.7	62.3	74.0	80.2	83.6	79.5	68.7	58.3	49.4	43.0		
		MCWB	37.8	37.6	39.7	50.0	59.4	62.5	66.1	64.2	56.7	50.2	45.1	38.3		
	2%	DB	39.0	39.2	43.1	55.8	67.9	76.8	78.9	75.0	63.5	53.7	45.4	40.5		
		MCWB	35.6	36.5	37.6	45.6	55.1	61.2	64.5	62.8	55.0	47.6	42.2	37.2		
	5%	DB	36.3	37.5	40.6	50.9	61.8	71.8	74.5	70.5	60.0	50.6	41.3	38.2		
		MCWB	34.2	35.1	36.8	42.6	51.8	59.2	62.6	60.2	53.1	45.9	38.6	35.6		
	10%	DB	34.3	35.3	38.4	46.9	56.7	66.3	70.6	66.5	56.6	47.7	38.5	36.2		
		MCWB	32.7	33.3	35.5	40.8	48.5	56.8	60.2	58.8	51.0	44.4	36.5	34.0		
	0.4%	WB	37.9	38.5	40.4	50.6	59.9	64.4	67.3	65.6	58.9	51.7	46.2	39.6		
		MCDB	41.0	40.3	46.0	62.0	72.4	76.8	80.0	75.8	65.4	56.8	49.9	41.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.1	37.1	38.8	46.1	55.8	62.0	65.4	63.5	56.2	48.3	42.5	37.5		
		MCDB	38.0	38.9	42.3	54.0	65.7	73.6	76.5	72.1	62.0	52.3	45.0	40.0		
	5%	WB	34.5	35.6	37.4	43.6	52.9	59.7	63.6	61.5	53.6	46.5	38.9	35.9		
		MCDB	36.2	37.1	40.2	49.8	60.9	70.7	72.9	68.8	59.3	50.1	40.9	37.8		
	10%	WB	32.6	33.0	35.7	41.5	49.5	57.3	61.4	59.7	51.4	44.5	36.6	34.3		
		MCDB	34.2	35.1	38.2	46.2	55.9	65.5	68.9	65.5	55.7	48.0	38.4	36.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.0	14.8	15.3	15.9	16.6	16.8	15.6	13.5	10.4	9.1	11.1		
		MCDBR	10.4	11.7	15.9	23.6	26.5	27.4	24.8	22.9	19.4	15.8	12.4	11.4		
		MCWBR	8.8	9.5	11.1	13.9	15.0	13.1	11.8	11.6	11.6	10.5	9.7	9.4		
		MCDBR	10.1	11.1	13.5	20.6	25.3	24.9	22.8	20.5	18.0	14.3	12.0	10.8		
	5% WB	MCWBR	8.8	9.8	9.7	12.7	14.8	12.5	11.4	11.2	11.2	10.2	9.6	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.199	0.260	0.291	0.322	0.332	0.330	0.341	0.330	0.303	0.285	0.209	N/A		
	taud		2.015	2.218	2.256	2.322	2.451	2.502	2.486	2.506	2.567	2.464	2.087	N/A		
	Ebn at Noon		138	218	257	273	280	283	276	270	257	212	131	N/A		
	Edh at Noon		9	18	27	31	30	30	30	26	20	15	8	N/A		
(44) All-Sky Solar Radiation	RadAvg		54	223	654	1238	1463	1551	1495	1132	672	288	81	26		
	RadStd		6	31	72	97	104	140	170	110	62	32	10	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page